

M5StickC BLE STRIDE Practice

Application Info

Minimal BLE-focused threat model for practice. Scope: Operator -> M5StickC -> BLE Gateway.

Architecture

Introduction

M5StickC device sends BLE alert payloads to a trusted gateway app in a conference environment.

Dataflow

Introduction

- Operator triggers event on M5StickC
- M5StickC sends BLE payload to gateway app
- Trust boundary: BLE radio environment is untrusted

Assumptions

Assumption Number	Assumption	Linked Threats	Linked Mitigations	Comments
A-0001	BLE secure pairing is enabled and only approved gateway devices are paired.	T-0001 : A threat actor with BLE proximity to the conference demo area can impersonate the trusted BLE gateway, which leads to unauthorized receipt of device payloads and possible relay of false data, resulting in reduced confidentiality and integrity of BLE		

Assumption Number	Assumption	Linked Threats	Linked Mitigations	Comments
		payload and device-gateway trust relationship		

Threats

Threat Number	Threat	Mitigations	Assumptions	Status	Priority	STRIDE	Comments
T-0001	A threat actor with BLE proximity to the conference demo area can impersonate the trusted BLE gateway, which leads to unauthorized receipt of device payloads and possible relay of false data, resulting in reduced confidentiality and integrity of BLE payload and device-gateway trust relationship	M-0001 : Require bonded BLE pairing and reject unpaired peers.	A-0001 : BLE secure pairing is enabled and only approved gateway devices are paired.	Identified	High	S	

Mitigations

Mitigation Number	Mitigation	Threats Mitigating	Assumptions	Status	Comments
M-0001	Require bonded BLE pairing and reject unpaired peers.	T-0001 : A threat actor with BLE proximity to the conference demo area can impersonate the trusted BLE gateway, which leads to unauthorized receipt of device payloads and possible relay of false data, resulting in reduced confidentiality and integrity of BLE payload and device-gateway trust relationship		Identified	

Impacted Assets

Assets Number	Asset	Related Threats
AS-0001	BLE payload	T-0001 : A threat actor with BLE proximity to the conference demo area can impersonate the trusted BLE gateway, which leads to unauthorized receipt of device payloads and possible relay of false data, resulting in reduced confidentiality and integrity of BLE payload and device-gateway trust relationship
AS-0002	device-gateway trust relationship	T-0001 : A threat actor with BLE proximity to the conference demo area can impersonate the trusted BLE gateway, which leads to unauthorized receipt of device payloads and possible relay of false data, resulting in reduced confidentiality and integrity of BLE payload and device-gateway trust relationship